

1. Sum of digits of a number using for loop?

```
num = 54321
sum_digits = 0
for digit in str(num):
    sum_digits += int(digit)
print("Sum:", sum_digits)
```

2. Find all prime numbers from 1 to 50

```
for num in range(2, 51):
    for i in range(2, num):
        if num % i == 0:
            break
    else:
        print(num, end=' ')
```

3. Print only even-position characters in a string

```
s = "PYTHONLOOP"
for i in range(0, len(s), 2):
    print(s[i], end="")
```

4. Print only non-repeating characters from a string

```
s = "programming"
for ch in s:
    if s.count(ch) == 1:
        print(ch, end=' ')
```

5. Check if a number is an Armstrong number (e.g. 153)

```
num = 153
sum = 0
for digit in str(num):
    sum += int(digit) ** len(str(num))
print("Armstrong" if sum == num else "Not Armstrong")
```

6. Find the largest digit in a number

```
num = 986532
max_digit = 0
for ch in str(num):
    if int(ch) > max_digit:
        max_digit = int(ch)
print("Max digit:", max_digit)
```

7. Print all palindromic numbers between 10 and 200

```
for i in range(10, 201):
    if str(i) == str(i)[::-1]:
        print(i, end=' ')
```

8. Count frequency of each character in a string (without collections module)

```
s = "loopinside"
for ch in sorted(set(s)):
    count = 0
    for c in s:
        if c == ch:
            count += 1
    print(f"{ch}: {count}")
```

9. Find factorial of each digit in a number and sum them

```
num = 145
sum = 0
for ch in str(num):
    fact = 1
    for i in range(1, int(ch)+1):
        fact *= i
    sum += fact
print("Sum of factorial of digits:", sum)
```

10. Count frequency of words in a sentence

```
sentence = "Python is fun and Python is easy"
words = sentence.split()
for word in set(words):
    count = 0
    for w in words:
        if w == word:
            count += 1
    print(f"{word}: {count}")
```

11. Print all pairs of numbers from 1 to 5

```
for i in range(1, 6):
    for j in range(1, 6):
        print(f"({i},{j})", end=' ')
    print()
```

12. Reverse each word in a sentence

```
sentence = "Python for loop practice"
words = sentence.split()
for word in words:
    rev = ""
    for i in range(len(word)-1, -1, -1):
        rev += word[i]
    print(rev, end=' ')
```

13. Find the longest word in a list

```
words = ["python", "list", "comprehension", "loop"]
longest = ""
for word in words:
    if len(word) > len(longest):
        longest = word
print("Longest word:", longest)
```

14. Remove duplicate characters from a string

```
s = "programming"
result = ""
for ch in s:
    if ch not in result:
        result += ch
print("Without duplicates:", result)
```

15. Find common elements in two lists

```
list1 = [1, 2, 3, 4]
list2 = [3, 4, 5, 6]
common = []
for i in list1:
    for j in list2:
        if i == j and i not in common:
            common.append(i)
print("Common elements:", common)
```

16. Print all unique words in a sentence

```
sentence = "python is fun and python is powerful"
words = sentence.split()
for i in range(len(words)):
    count = 0
    for j in range(len(words)):
        if words[i] == words[j]:
            count += 1
    if count == 1:
        print(words[i], end=' ')
```

17. Find all pairs in a list that sum to a given number

```
nums = [1, 3, 5, 2, 4, 6]
target = 7
for i in range(len(nums)):
    for j in range(i+1, len(nums)):
        if nums[i] + nums[j] == target:
            print(f"{nums[i]} + {nums[j]} = {target}")
```

18. Remove duplicates from a list (without set)

```
nums = [1, 2, 3, 2, 4, 3, 5]
unique = []
for num in nums:
    if num not in unique:
        unique.append(num)
print("Without duplicates:", unique)
```